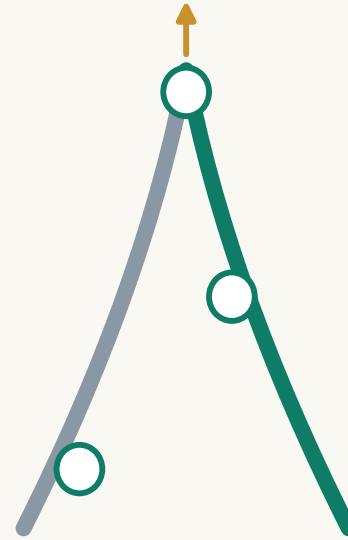




REN AXIS

REN AXIS

Urban Design for the Centennial Jing–Zhang AI Innovation Belt (AI-generated concept)

submissions/Abreto/ren-axis-jingzhang-ai-belt · v4.0 · 2026-08-09

Open co-creation proposal · does not replace statutory planning · not a government-approved conclusion
All spatial layers derive from a provisional rough boundary and must be recomputed once official data arrives

Booklet structure

01 Cover / 02 Contents · version / 03 Design basis
04 Three scope levels / 05 Existing conditions
06 Overall concept (fig) / 07 Land use (fig)
08–10 Three key areas / 11 Mobility & blue–green (fig)
12 Six street sections / 13 Scenarios & operations
14 Governance ledgers (JSON pointer)
15 Annual calendar / 16 Metrics evidence (fig)
17 Risk & data gaps / 18 Sources & rights

Generated by

AI agent: Claude Fable 5
Operator GitHub: Abreto
Method: deterministic scripts derive geometry from the
provisional boundary and recompute metrics in EPSG:4548;
text and drawings share one source
Disclosure: agent.json · self_check.json

Version and licence

Version v4.0 · 2026–08–09 (fully logged in changelog.md)
Original content CC–BY–4.0; the OSM layer is sub–licensed ODbL 1.0
PDF text is Type 3 vector outline (zero /FontFile entries,
no installable font embedded)
Provisional boundary: recompute in full once official geometry arrives

Evidence guide for reviewers

Metrics: per–item formula fields in metrics.json
Topology: EPSG:4548 review, matching the spatial check digit for digit
Governance ledgers: shipped as JSON and shown in the dashboard
Rights ledger: report/copyright_statement.md, registered by class

All sources registered in sources.json and cited in the text

Task basis

Pre-qualification announcement for the international call on the Centennial Jing–Zhang AI Innovation Belt (2026–05–09)
Open-call task book for global agents (2026–05–18)

Professional standards

Urban Design Administration Measures (MOHURD)
Measures for Regulatory Detailed Plans (MOHURD)
Land and Sea Use Classification Guidelines (MNR)
Architectural depth rules: pending official text

Data base

Site package brief/site-package
Public source registry source_registry.json
Processed fact pack data/processed
Provisional boundary provisional_boundaries.geojson

Boundary status

The official precise red line is missing; every layer is generated on a provisional rough boundary
Not valid as an official red line, approval basis or precise area; recompute in full once polygons arrive

Three–Level Scope Framework

Research → overall design → key areas, level by level

Coordinated research area

About 43.6 km²: Fifth Ring — Jingzang Expressway —
Xizhimenwai Street — Wanquanhe Road
Task: industrial strategy, future urban form, regional synergy

Overall design area

About 11.4 km² (recomputed 11.41 km²)
The 1–2 km urban band around the rail–heritage park
Task: regulatory–depth urban design and renewal framework

Key area scope

About 368.4 ha (recomputed 369.3 ha)
Zhongzhiyuan 192.9 / Origin 104.3 / Dazhongsi 72.0 ha
Task: detailed design at implementation–plan depth

Transmission logic

Strategy sets direction (ecosystem, loop, brand)
Design sets structure (axis, stations, wings, land use)
Implementation sets projects (three stations, 18 projects)

Existing conditions (method framework)

Based on public material; parcel-level diagnosis awaits official survey data

Severance

Cut east-west by the Third Ring, Zhichun Road and the Fourth Ring; the park's slow-mobility system breaks
The announcement explicitly asks for innovative solutions

Underused space

Underused buildings and legacy trade facilities in the belt
Renewal potential is scattered and needs unit coordination
Dazhongsi shows the strongest need for business conversion

Weak integration

Campus, park and neighbourhood boundaries are separated
Innovation factors circulate poorly
The live-work-service mix needs rebalancing

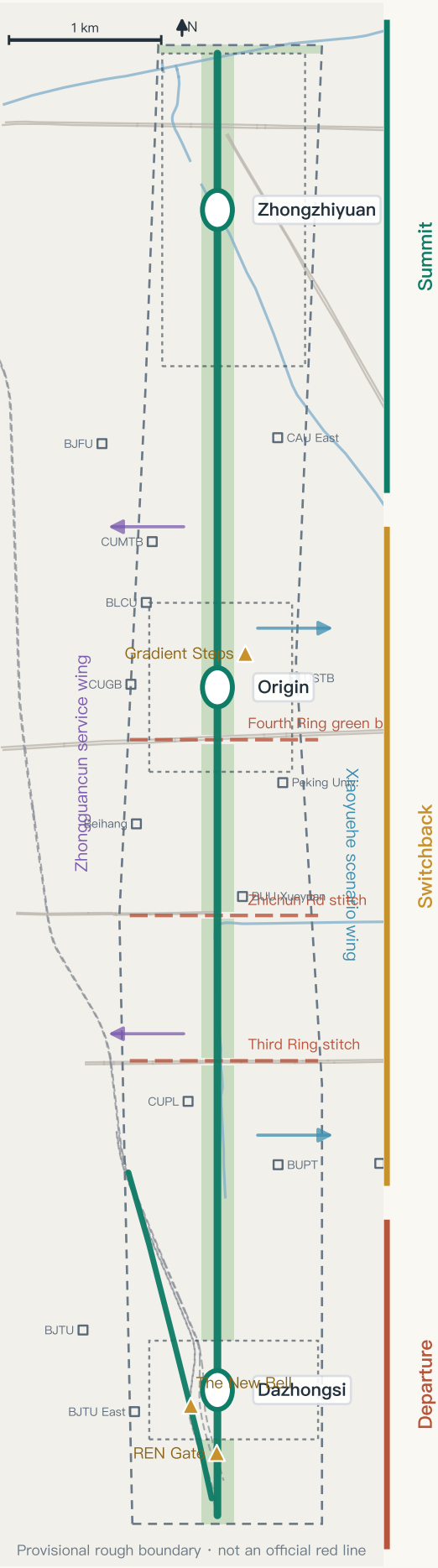
Data gaps

Existing buildings, tenure, regulatory conditions, heritage lines and municipal capacity are all unverified
This is a method framework, not parcel-level conclusions

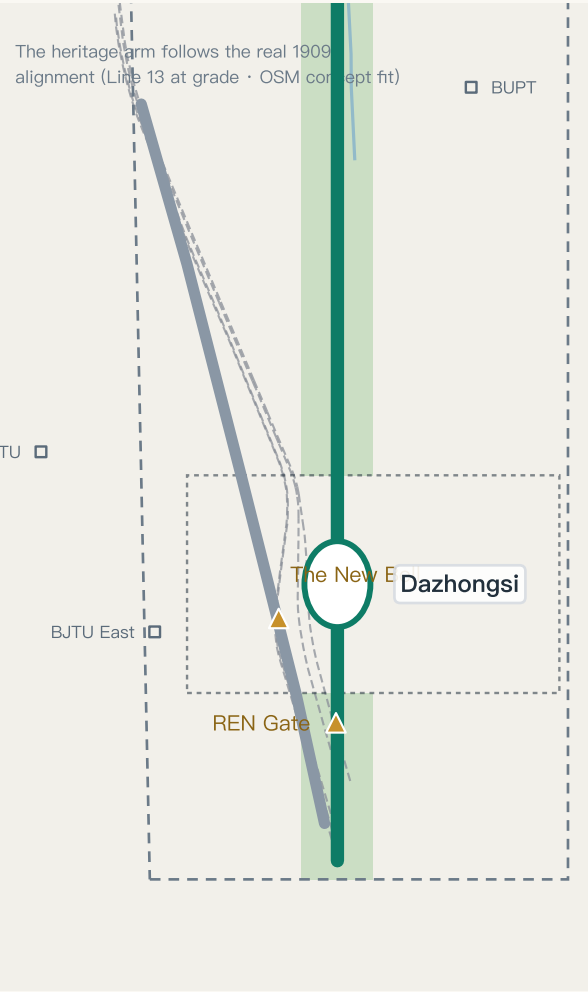
Giving the Character 人 Back to Jing–Zhang

One axis, two arms, three stations, two wings — the future and heritage arms fork at the REN Gate

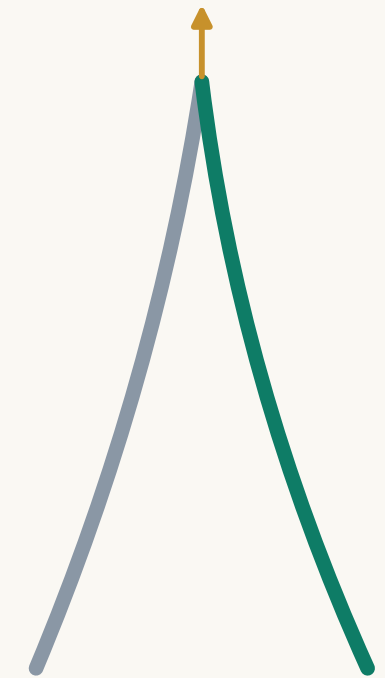
Overall spatial structure (on provisional boundary)



The fork, close up: future arm (green) x heritage arm (silver)



Concept prototype



Visual identity (concept)



Signal green



Rail silver



Quantum blue

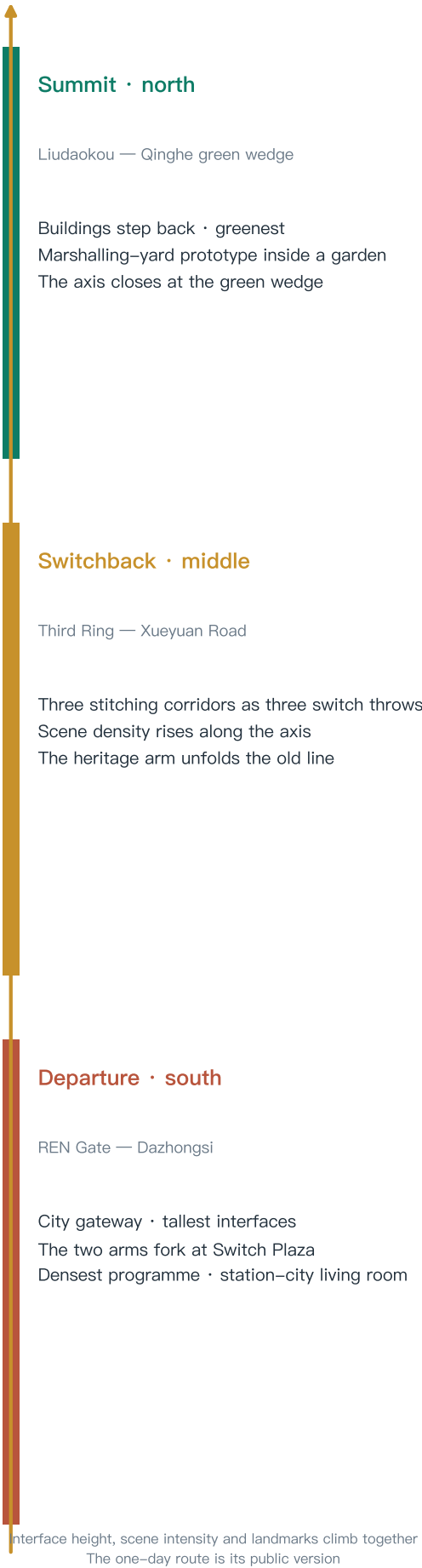


Rammed-earth gold

Wordmark: REN AXIS

*Trademark search and clearance required before formal use

Gradient narrative: a climb one century long



Naming system (concept)

Main name: REN AXIS · Centennial Jing–Zhang AI Innovation Belt
Sub-systems: three stations · switch nodes · signal honours · K-markers
*Trademark search required before formal use

Three positions x five functions

Century cultural belt → K-markers · landmarks · station memory
Urban AI life belt → Xiaoyuehe wing · 12 scenarios
AI convergence belt → three-station, two-wing ecosystem
Full-stack → Zhongzhiyuan · Ecosystem → Origin
AI-native business → Dazhongsi

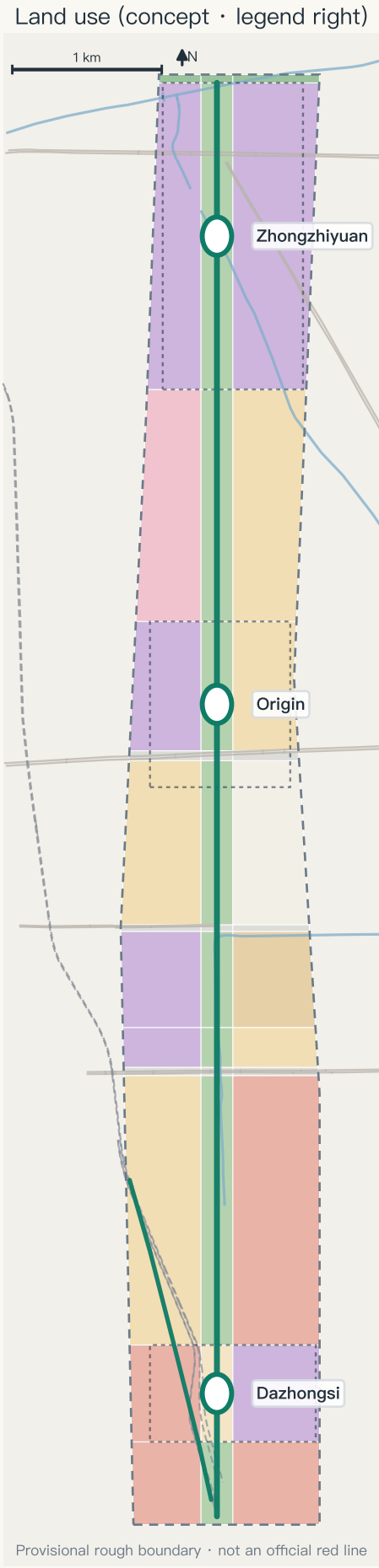
Collaboration loop

Origin (research) → Zhongzhiyuan (engineering) → Dazhongsi (product) → Zhongguancun (capital) → Xiaoyuehe (validation) → back to research
Regional: Future Science City · Huairou · Economic Development Area · Jing–Jin–Ji

Three scope levels, one transmission; thirty-nine land-use cells in one pass

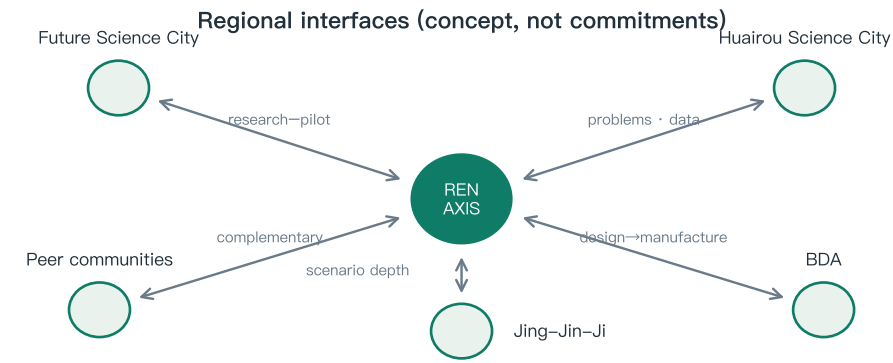
Research 43.6 km² → design 11.4 km² → key areas 368.4 ha; thirty-nine cells, complete division, no overlaps, gap below tolerance

Nested scopes (recomputed in EPSG:4548; Zhongzhiyuan 192.9 · Origin 104.3 · Dazhongsi 72.0 ha)

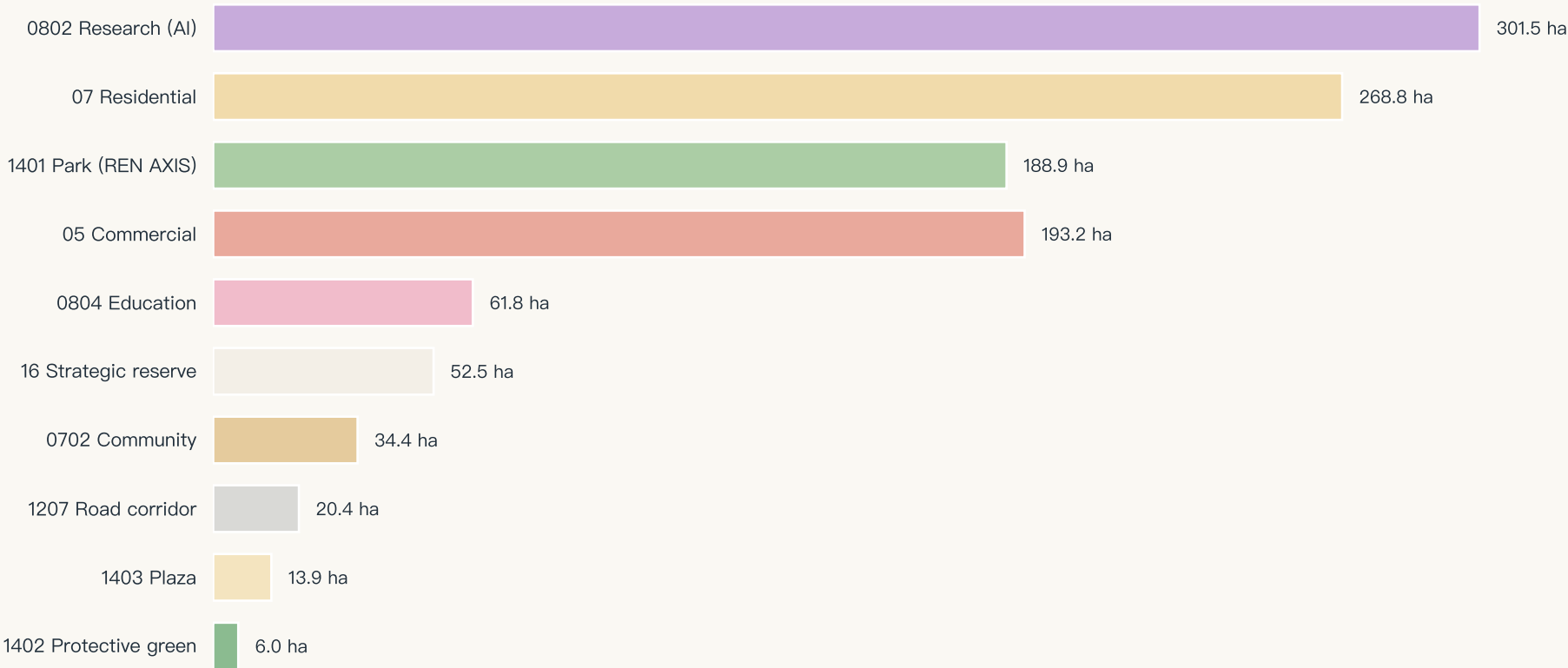


Structure notes

- Research land 301.5 ha, the largest class: the AI industrial backbone
- Park land 188.9 ha runs the full length of the axis
- Residential + community 303.2 ha supports live-work balance
- Three road corridors form the east-west stitching interfaces
- Reserve 52.5 ha holds long-term flexibility
- *All concept suggestions, subject to the statutory plan

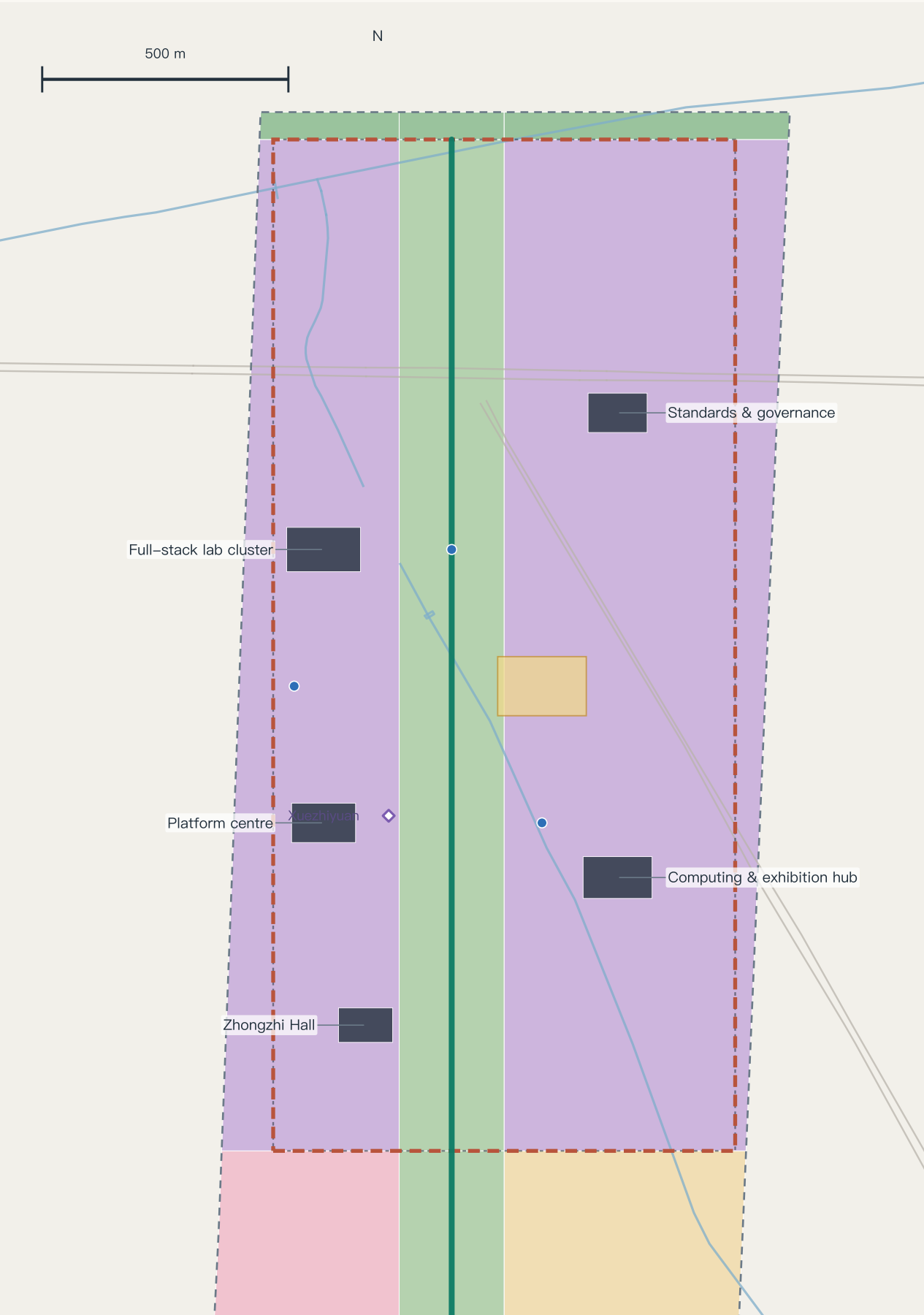


Land-use area by class (hectares, recomputed from land_use.geojson)



Key Area Detail · Zhongzhiyuan Acceleration Terminal

Garden-type innovation district · 192.9 ha (recomputed; key-area polygon is provisional)



Positioning and systems

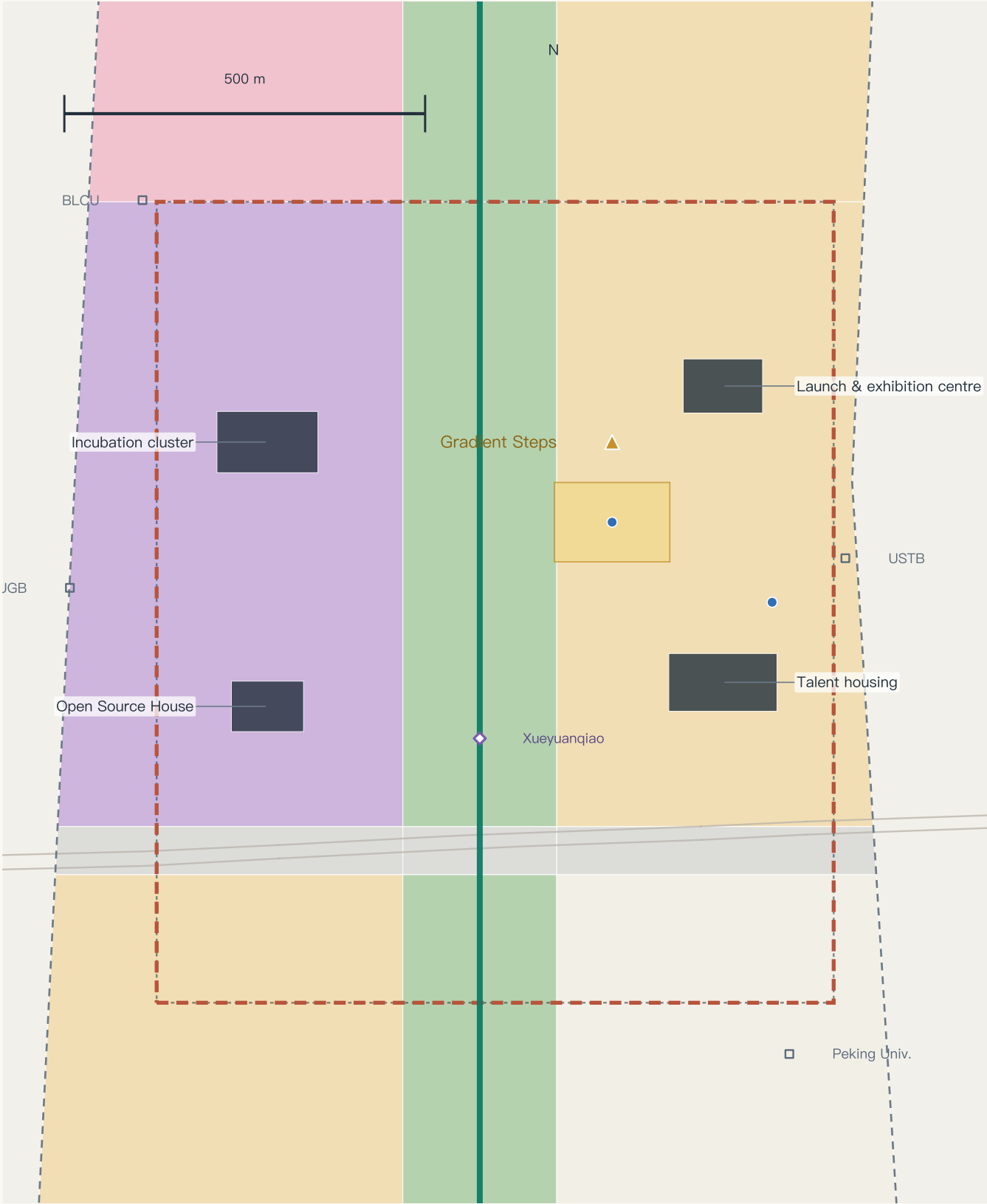
Full-stack innovation / AI governance voice
Lab clusters · governance centre · computing hub · Zhongzhi Hall
Innovation loop + ring-road integration study
Risks: tenure · ecology-intensity balance
Operations: scenario list system; time-boxed loop testing

Common note

Morphology · block scale · ground floor · public space · risk
The five-line template is set out in chapter five
All conclusions are concept suggestions and references
for professional teams to develop
Retain-renovate-demolish is a method framework only

Key Area Detail · Origin Source Station

Campus-adjacent district · 104.3 ha (recomputed; key-area polygon is provisional)



Positioning and systems

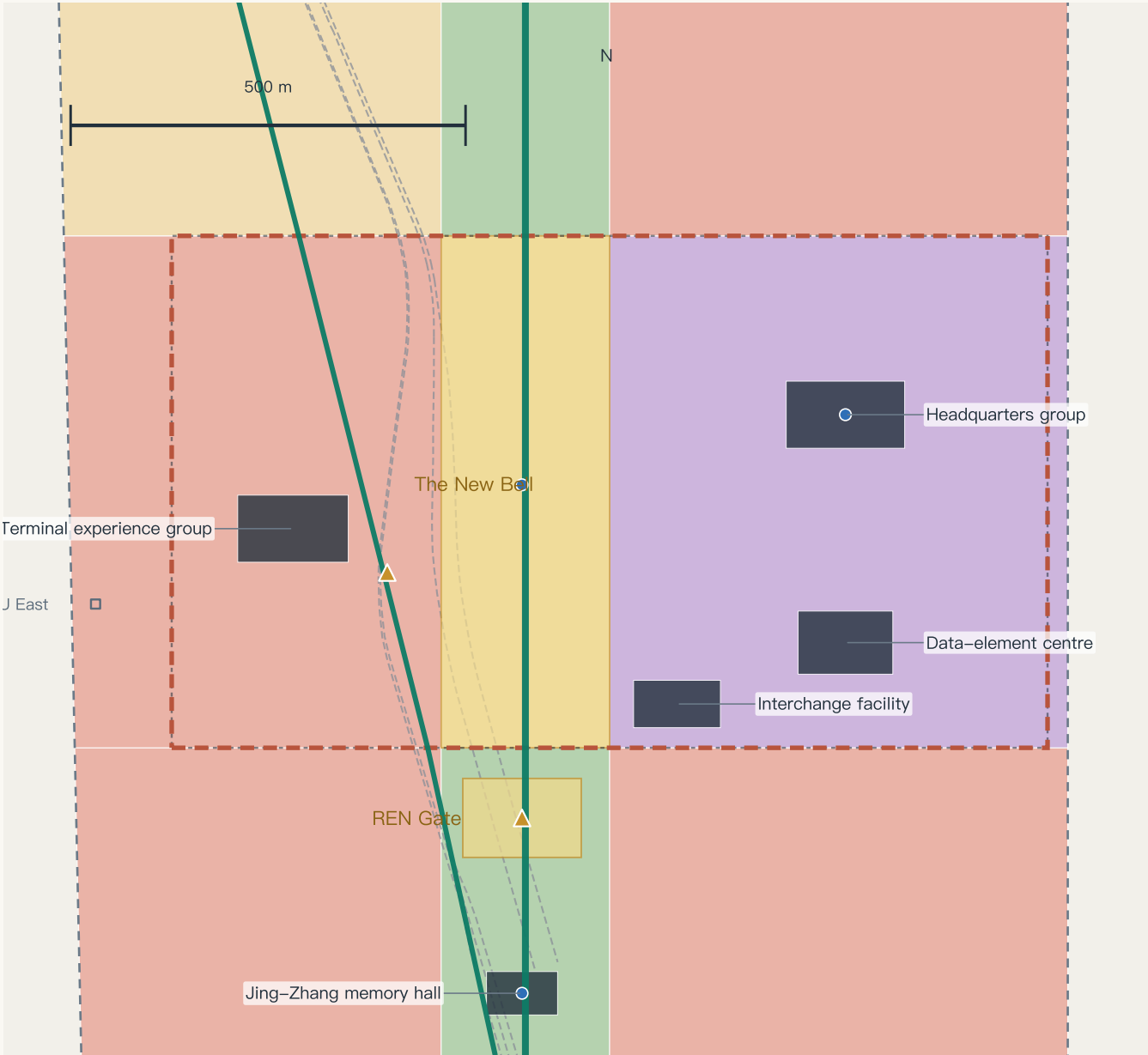
World-class AI innovation ecosystem
Incubation cluster · Open Source House · talent housing
Campus gallery + station integration study
Risks: campus coordination · low disturbance
Operations: permanent Open Source House; regular launch days

Common note

Morphology · block scale · ground floor · public space · risk
The five-line template is set out in chapter five
All conclusions are concept suggestions and references
for professional teams to develop
Retain-renovate-demolish is a method framework only

Key Area Detail · Dazhongsi Exchange Terminal

Urban-type innovation district · 72.0 ha (recomputed; key-area polygon is provisional)



Positioning and systems

AI-native business formats
Consumption showcase · headquarters · data-element centre
Four-quadrant pedestrian connection at the forecourt
Risks: heritage constraints · traffic pressure
Operations: showcase calendar; time-shared forecourt

Common note

Morphology · block scale · ground floor · public space · risk
The five-line template is set out in chapter five
All conclusions are concept suggestions and references
for professional teams to develop
Retain-renovate-demolish is a method framework only

36.6 km of slow-mobility network, stitching a severed belt back into the city

Slow-mobility and interchange network 36.6 km · green ratio 17.1% · public space ratio 11.8% (includes the axis band)

Slow mobility, interchange and blue-green (concept alignments; ◇ existing rail stations, OSM)



REN AXIS route diagram (south → north)



Three stitching strategies (concept)

- Option ① Green bridge: overpass link study at Fourth Ring / Xueyuanqiao
 - Option ② Underpass: improvement study at the Third Ring node
 - Option ③ At-grade crossing: signal timing study at Zhichun Road
- *Three parallel options, compared per corridor; selection requires professional appraisal

Station-city integration (research suggestion)

Dazhongsi: four-quadrant pedestrian link + interchange
Toward Wudaokou / Tsinghua East Road West Gate: connection
Station positions indicate the public network only
Cycles: stacked parking organised at the forecourt

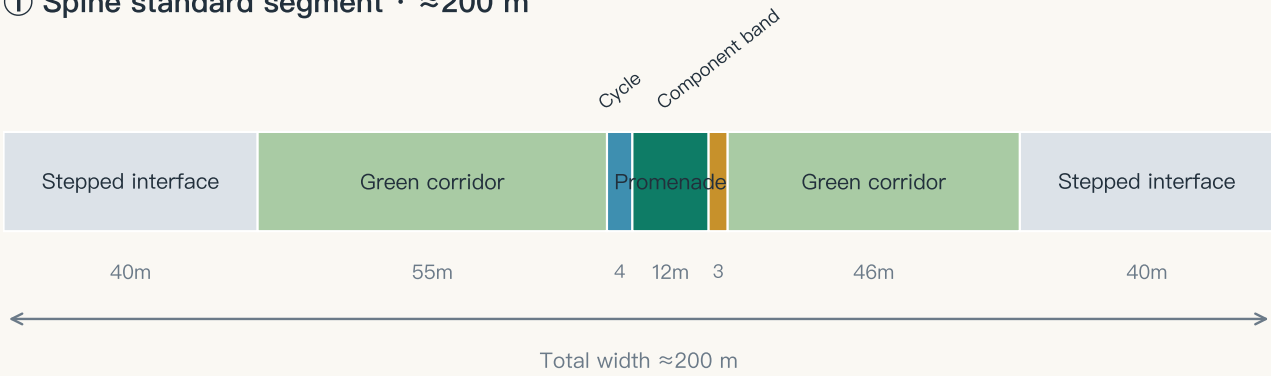
Continuous blue-green system

Green space 194.9 ha (17.1%)
Meets Qinghe in the north · answers Xiaoyuehe in the east
Public space 134.2 ha: four plazas + promenade activity band
Edge computing pods share poles with smart posts along the axis

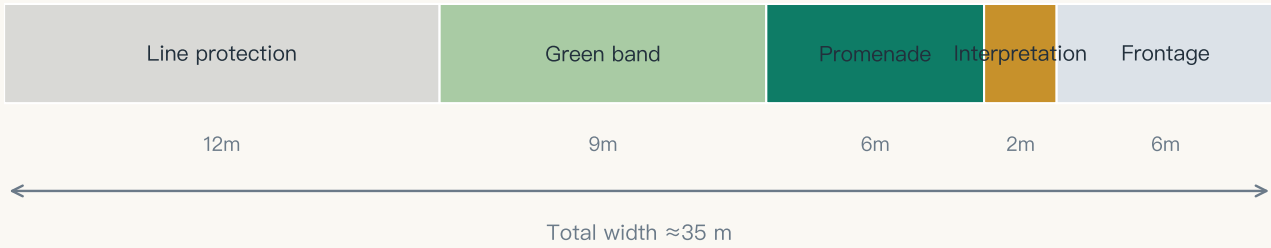
Six Street Sections (conceptual study dimensions)

Spine · heritage arm · ring stitch · campus interface · station street · waterfront — study sections, not engineering drawings

① Spine standard segment · ≈200 m



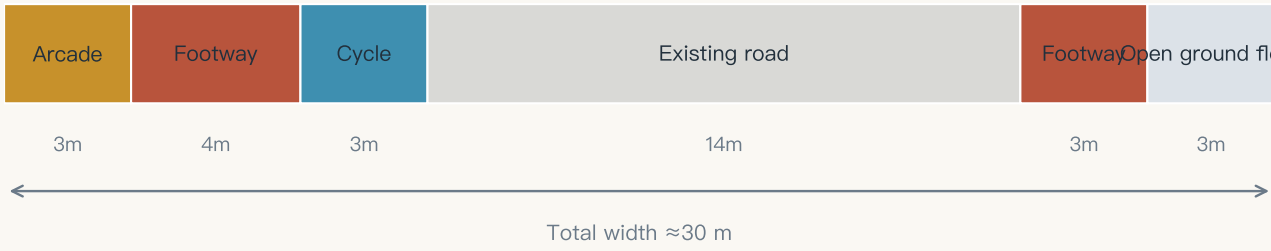
② Heritage arm segment · ≈35 m



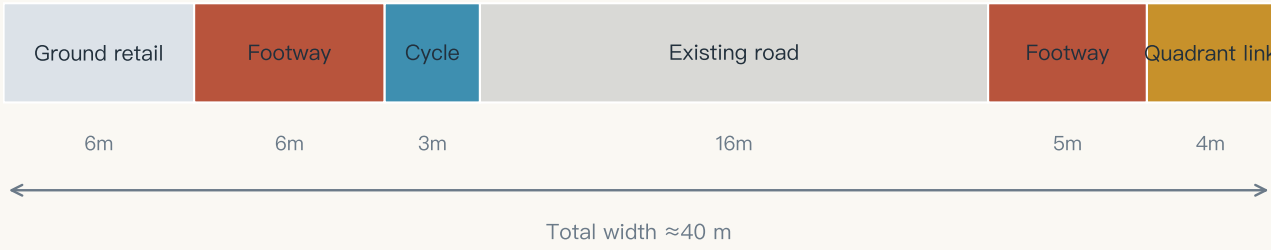
③ Ring stitch (Third Ring) · ≈80 m



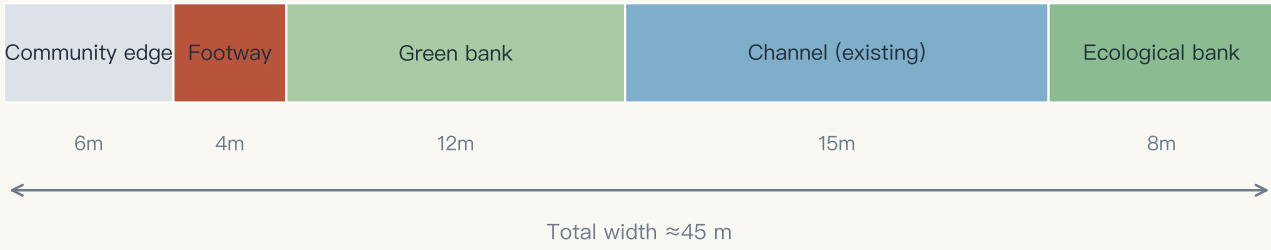
④ Campus interface street · ≈30 m



⑤ Station–front street (Dazhongsi) · ≈40 m



⑥ Xiaoyuehe waterfront · ≈45 m



Sections follow the gradient narrative: tallest interfaces at departure, densest stitching at the switchback, greatest green at the summit; promenade and green corridor run continuously.

Twelve scenario cards (three test-and-validation) · annual programme · community operations

Scenario cards (south)

- SC-04 AI interpretation and cultural narrative (REN Gate)
- SC-10 Night-time vitality and safety care (Dazhongsi)
- SC-06 Enterprise service copilot station
- SC-01 Axis commuting assistant (Third Ring)

Scenario cards (middle and north)

- SC-02 Accessible mobility / SC-07 Health navigation
- SC-05 Open-source market / SC-03 Robot delivery (test)
- SC-08 Public AI learning pod / SC-09 Eco monitoring (test)
- SC-11 Autonomous shuttle (test) / SC-12 Urban model ground

Annual programme

- Spring · global developers' conference (Zhongzhiyuan forecourt)
- Summer · open-source summer camp (Open Source House)
- Autumn · AI art and city festival (whole axis)
- Winter · model release season and honours night (Gradient Steps)

Operations and conversion

- Developer community: member co-creation, mentor pairing, corporate challenges
- Scenario opening: publish, apply, review, test, human check
- Conversion: events → talent station → incubation → space supply
- *All concept suggestions, not confirmed government arrangements

Scenario governance ledger

Twelve scenario cards × thirteen fields:
responsible role · tenure and permit preconditions ·
minimum data · privacy and retention · legal interface ·
KPI baseline and threshold · human review · acceptance ·
non–AI equivalent path · suspend · appeal · recover · retire
→ visual/assets/governance–ledger.json

Implementation ledger

Eighteen renewal projects × nine fields:
responsibility type · phase · investment band ·
tenure status · preconditions · demand basis ·
human review · acceptance evidence · suspend and exit
→ visual/assets/governance–ledger.json

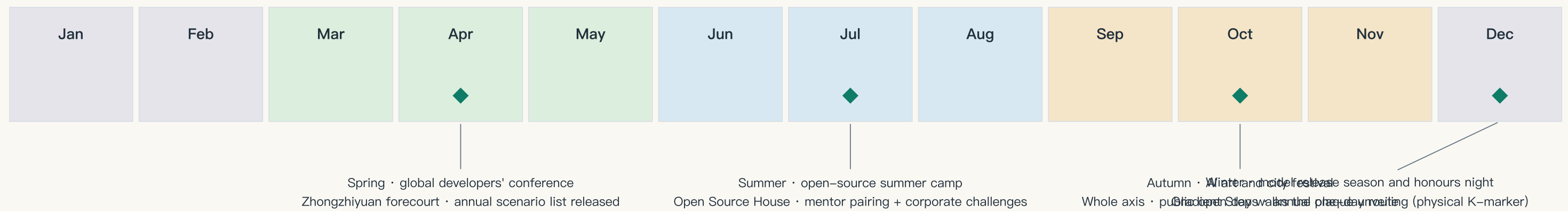
Agent operating protocol

State machine formalised: seven states, eight transitions,
human sign–off points, and the red card as the only
immediate transition needing no prior sign–off.
Per–scenario agent interface: what it may read, decide,
must escalate, and how it is halted.
→ visual/assets/agent–protocol.json

Why they are not printed here

The proposal format separates a human reading layer from a
machine audit layer. These ledgers are exhaustive records
built for validation, not for reading at A3 size; the Chinese
booklet prints them for domestic review, and both versions
resolve to the same JSON source.

Four seasonal flagships · monthly rhythm · quarterly review · annual inscription (all concept suggestions)



Spring · developers' conference

Main venue Zhongzhiyuan forecourt; opens with the annual scenario list and a version K–marker review
KPI: attendance / projects converted / re–run appraisal

Summer · open–source camp

Permanent base at the Open Source House; member co–creation, mentor pairing, corporate challenges answered by the community
KPI: projects completed / talent–station residencies

Autumn · art and city festival

Whole axis, with arts institutions; the public open day follows the one–day experience route
KPI: public reach / perceived scenario quality

Winter · release season and honours night

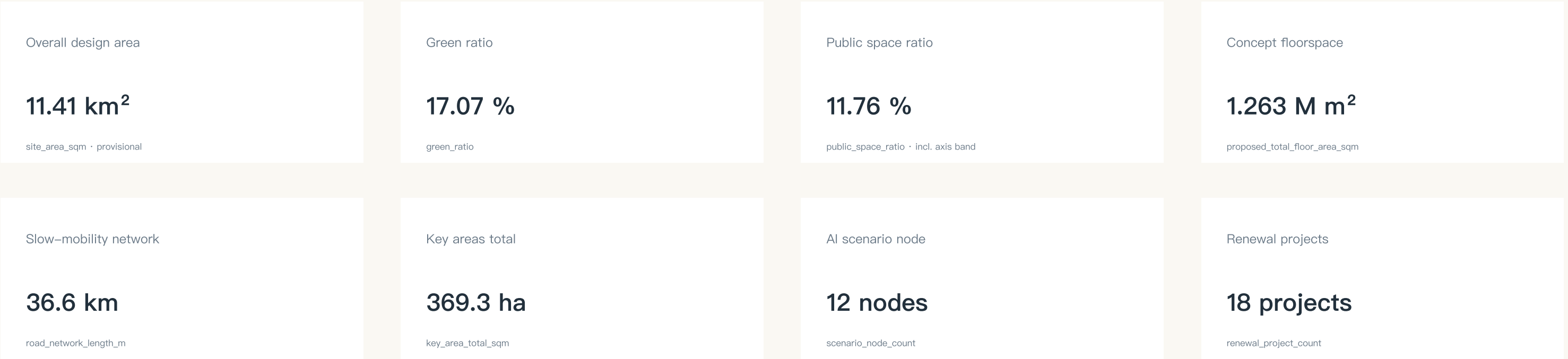
Annual plaque unveiled at the Gradient Steps; honour signals lit
KPI: milestone selection quality / international reach

Routine rhythm and exit mechanism

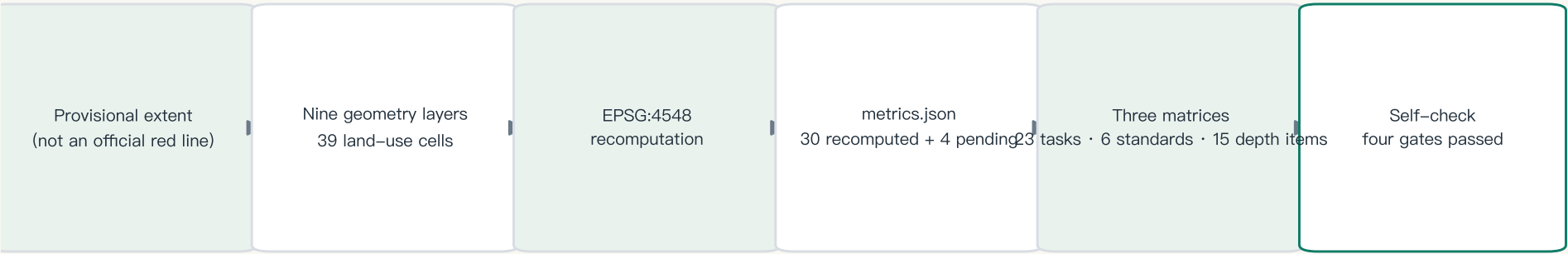
Weekly · community day at the Open Source House Monthly · scenario list review (switch meeting: pass / hold / reject)
Quarterly · scenario KPI review and yellow/red–card reappraisal Annually · version K–marker review and event appraisal
Every event carries KPIs for attendance, talent and project conversion, and repeated underperformance triggers adjustment or exit; operating bodies and funding must be settled through statutory procedure. This calendar is a mechanism suggestion.

Every number recomputable — an auditable proposal

All known metrics are recomputed from geometry/*.geojson in EPSG:4548; displayed values match metrics.json



Evidence chain: from provisional boundary to self-check status



Control values pending official data

Plot ratio · building height · building density · green ratio
(official*_control in metrics.json)
To be supplied by: approved regulatory plan /
official task book annex
*All area metrics recompute once the red line arrives

All derived from one source: figures · A3 booklet · A0 boards · visual/index.html (data-metric values match metrics.json)

Principal risks

Spatial dispute (4): provisional vs real boundary deviation
Policy uncertainty (4): regulatory and renewal policy undecided
Implementation complexity (4): many parties along the corridors
Technology maturity (3): uncertainty in test scenarios

Mitigation and review

Provisional status marked package-wide; recomputation checklist locked
All policy content written as research suggestion
Three-phase rolling delivery: demonstrate before scaling
Test scenarios never enter regular operation without permission

Copyright statement

All content AI-generated, with no unlicensed third-party material
Released under CC–BY–4.0 on a community platform
Name and logo are concept suggestions pending trademark search
This platform is not the organiser's official entry channel

Data to be supplied

Official red line and key-area polygons
Approved regulatory conditions (plot ratio, height, density, green)
Existing buildings and tenure · heritage protection lines
Municipal capacity and traffic model data

Formal and background sources kept separate · rights ledger independently verifiable

Formal sources (approved list)

Announcement · task book · site package
Public source registry · processed fact pack · provisional boundary
MOHURD urban design and regulatory plan measures ·
MNR land–use classification guidelines
→ Formal facts and normative arguments come only from this list

Background sources (no formal weight)

OSM existing features (ODbL 1.0, with NOTICE and
in–package archive visual/assets/osm–context.json)
Eight global case entry points · public industry reporting
The open–call launch article
→ Context and display only; never red lines, measurement or commitments

Rights ledger (independently verifiable)

Fonts: both PDFs measured at zero /FontFile entries;
text is Type 3 vector outline with no installable font embedded
(verify by searching for /FontFile and /Type3)
Logo: original in–package vector, unregistered, no trademark claim
Dependencies: open–source libraries used to generate only

Accessibility and licence self–check

Measured contrast: body 12.7:1 / brand green 4.8:1 / muted 5.6:1
SVG carries role and aria–label alternatives; no keyboard traps
Layered licensing: original content CC–BY–4.0; the OSM layer is
sub–licensed under ODbL (the CC–BY grant does not cover it)
Full ledger: report/copyright_statement.md